Planting trees in graphs, and finding them back

Ludovic Stephan[1,2], Laurent Massoulié[1,3], Don Towsley[4]

[1]: Inria - Ecole Normale Supérieure - PSL Research University, [2]: Sorbonne Université, [3]: Microsoft Research Inria Joint Centre, [4]: University of Massachusetts Amherst

Background

Inference problems: task of detecting/finding a **planted structure** inside random data (usually random graphs).

Two main questions:

- **Detection**: is there a planted structure inside the given data ?
- **Reconstruction**: given the existence of a planted structure, where is it ?

Common examples: stochastic block model, planted clique...

Usual behaviour in inference problems :

- a phase where detection/reconstruction are **information-theoretically impossible** when the signal is too low
- a phase where detection and reconstruction are possible but **computationally hard** (no polynomial algorithm known)
- finally, a phase where both are easy: polynomial algorithms known

Main questions :

Is this the case for all such problems ? What happens inside the computationally hard phase ?

Setting

Similar to the planted clique problem: **planted subgraph** inside a random graph. However, we consider a sparse graph $\to$ need a sparse subgraph !

Model definition: We consider a random graph G, generated as follows:

- Under the null hypothesis H_0, G is drawn under $\mathcal{G}(n, \lambda/n)$;
- under the alternative hypothesis H_1, G is the union of a $\mathcal{G}(n, \lambda/n)$ random graph with a fixed graph Γ covering a random set of vertices $\mathcal{K}$ of size K.

The two questions become:

- **Detection**: is there a test that can separate H_0 from H_1 w.h.p ?
- **Reconstruction**: under H_1, how well can we reconstruct $\mathcal{K}$?

Note: it is not immediate that reconstruction is harder than detection !

We will focus on two graphs: a **path** of length K and a **complete D-ary tree** of height h.

Goal :

Understand the phase transitions for this particular inference problem

Proof methods

Edge counting: detection is always possible whenever $K = \omega(\sqrt{n})$.

Main detection method: Let

$$X_\Gamma = \#\text{copies of } \Gamma \text{ in } G$$

Then:

- if $\mathbb{E}[X_\Gamma] = o(1)$, easy detection !
- the likelihood ratio between H_1 and H_0 is

$$L = \frac{\mathbb{P}_1}{\mathbb{P}_0} = \frac{X_\Gamma}{\mathbb{E}[X_\Gamma]}$$

Therefore, if $\mathbb{E}[X_\Gamma^2] \sim \mathbb{E}[X_\Gamma]^2$, then $\mathbb{E}[L^2] \to 1$ and thus

$$d_{\mathrm{var}}(\mathbb{P}_0, \mathbb{P}_1) = o(1)$$

This implies that detection is **IT impossible** !

Main takeaway:

Detection $\iff$ study of X_Γ

Results for planted path

Dependency on λ: if the graph is subcritical ($\lambda < 1$), no path of length $\omega(\log(n))$ in the graph, but when critical ($\lambda > 1$), path of length $\Omega(n)$!

$\Rightarrow$ need to condition on λ

$\lambda < 1$: $\mathbb{E}[X_\Gamma] = o(1)$ whenever $K > \ln(n)/\ln(1/\lambda)$.

In this case, detection and reconstruction are easy, while they are impossible otherwise.

$K < \ln(n)/\ln(1/\lambda)$	$\ln(n)/\ln(1/\lambda) < K \ll n/\ln(n)$
Detection & reconstruction IT impossible	Detection & reconstruction easy

Summary of results for $\lambda < 1$

$\lambda > 1$: more interesting picture !

Markov chain argument to estimate $\mathbb{E}[X_\Gamma^2]$: detection impossible if $K = o(\sqrt{n})$, easy when $K = \omega(\sqrt{n})$.

However: reconstruction impossible whenever $K = o(n)$!

$K \ll \sqrt{n}$	$\sqrt{n} \ll K \ll n$
Detection & reconstruction IT impossible	Detection easy reconstruction IT impossible

Summary of results for $\lambda > 1$

Results for planted tree

Similarly to the planted path, the thresholds for detection and reconstruction depend on the value of λ ; however, the critical value λ_D is now that for the emergence of the **D-core** in a random graph.

$\lambda < \lambda_D$: the phase transition for detection happens at a height $h_* = \ln\ln(n)/\ln(D) + O(1)$. More specifically:

- if $h \geq h_* + O(1)$, detection is easy
- if $h \leq h_* - \ln(h_*)$, both detection and reconstruction are impossible.

However, **complete reconstruction is impossible** as long as $K = o(n)$: the leaves of the tree make up a positive proportion of the vertices, and are impossible to retrieve accurately.

$h < \ln\ln(n)/\ln(D)$	$\ln\ln(n)/\ln(D) < h < \ln(n)$
Detection & reconstruction IT impossible	Detection easy complete reconstruction impossible

Summary of results for $\lambda < \lambda_D$

$\lambda > \lambda_D$: not much more is known than the preliminary results ; detection is unknown when $K = O(\sqrt{n})$, and reconstruction is still impossible whenever $K = o(n)$.

Discussion

The phase transitions for this problem are substantially different from the usual ones ; namely:

- no computationally hard phase: detection/reconstruction are immediately easy
- detection and reconstruction are **not equivalent**, unlike in the closely related planted clique problem !

This can help foster new insights on inference problems, especially those whose planted structure is not close to a rank-one mtrix.

Main references

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